

EXERCISES AND EXACT SOLUTIONS

Beds, Staff, Supplies: Forecasting Service Capacity

Independent practice, facilitated discussion, and assessment reference

Production Unit	PU-B07-C22
Notebook	V7_B_N02
Version	0.1
Assessment boundary	Learning evidence—not operational authorization

Instructions

Complete Part A before consulting the solutions. State assumptions explicitly. Where evidence is insufficient, write an abstention or escalation response rather than inventing precision.

Exercise 1

Use rolling-origin evaluation.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 2

Add separate emergency and elective demand.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 3

Model staff-hours as a constraint.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 4

Explain why maximizing occupancy can be unsafe.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exact solutions and scoring guide

Solution 1

Reforecast from successive dates using only available history and score each horizon.

Solution 2

Use distinct concepts, arrival processes, and intervention rules; do not aggregate incompatible demand.

Solution 3

Convert workload through documented skill-mix and hours assumptions, then optimize under labour rules.

Solution 4

High occupancy removes surge buffer, increases delay and infection/quality risks, and ignores case mix and staffing.

Scoring rubric

Criterion	Strong evidence	Revision required
Decision framing	Owner, action, horizon, authority named	Model output substituted for decision
Evidence	Concepts, grain, provenance, limits explicit	Data treated as self-explanatory
Uncertainty	Abstention/escalation used appropriately	False precision or hidden limitations
Governance	Human authority and safeguards retained	Authority silently delegated